VIETNAM GENERAL CONFEDERATION OF LABOR

TON DUC THANG UNIVERSITY

FACULTY OF INFORMATION TECHNOLOGY



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FINAL REPORT

LINEAR ALGEBRA FOR IT

HO CHI MINH CITY, 2024

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Advised by
Ms. PHAM KIM THUY

HO CHI MINH CITY, 2024

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Ho Chi Minh city,27th December 2024.

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DECLARATION OF AUTHORSHIP

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Ho Chi Minh city, 27th December 2024

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CHAPTER 1. INTRODUCTION

1.1 Question 1

Investigate in-detail the singular value decomposition (SVD) of a rectangular matrix:

- Definition/Formulas of singular values
- At least one proposition related to singular values
- Definition/formulas of SVD
- At least one theorem related to a SVD finding
- Steps for finding a SVD
- Five applications of SVD
- Take to examples of SVD finding (examples from Internet/ books are not allowed) :
    - One with a matrix of size 2 rows $\times$ 3 columns and the rank of the matrix is 2
    - One with a matrix of size 2 rows $\times$ 3 columns and the rank of the matrix is 1

1.2 Question 2

Find a matrix P that diagonalizes $A = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$

1.3 Question 3

Use the Gram-Schmidt orthonormalization process to transform the basis $S = \{u_1 = (4; -8; 8), u_2 = (8; 8; 4), u_3 = (8; -4; -8)\}$ for $\mathbb{R}^3$ into an orthonormal basis.

1.4 Question 4

Consider the vector space $\mathbb{R}^3$ with two bases:

$\varepsilon = \{ \varepsilon_1 , \varepsilon_2 , \varepsilon_3 \}$ in which $\varepsilon_1 = (1, 0, 0),\ \varepsilon_2 = (0, 1, 0), \varepsilon_3 = (0, 0, 1)$

$\theta = \{ \theta_1 , \theta_2 , \theta_3 \}$ in which $\theta_1 = (1, 1, 0),\ \theta_2 = (0, 1, 1),\ \theta_3 = (1, 0, 1)$

a) Find the transition matrix from the basis ε to the basis θ.

b) Find the transition matrix from the basis θ to the basis ε.

CHAPTER 2. THEORETICAL BASIS

2.1 Question 1

=>Definition/Formulas of singular values:

-Singular values are fundamental quantities in linear algebra associated with a matrix A. For an m × n matrix A, the singular values are defined as the square roots of the eigenvalues of the symmetric matrix $A^T \cdot A$, which is n × n. These eigenvalues are guaranteed to be non-negative since $A^T \cdot A$ is positive semi-definite.

-The singular values, denoted $\sigma_1, \sigma_2, \dots, \sigma_r$ represent the magnitudes of the semi-axes of an ellipsoid resulting from the transformation by A.

-They are ordered as: $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r \geq 0$, where r = rank(A). Singular values describe how A scales or compresses along certain directions and are crucial for understanding the geometry of transformation.

=> At least one proposition related to singular values:

-For any m × n matrix A, the singular values $\sigma_1, \sigma_2, \dots, \sigma_r$ [where r = min (m, n)] satisfy the following proposition:

-Orthogonality and Singular Values: The columns of the matrix U (from $A = U\Sigma V^T$) are orthogonal eigenvectors of $A \cdot A^T$, and the columns of V are orthogonal eigenvectors of $A^T \cdot A$. The singular values σ_i are the square roots of the eigenvalues of both $A^T \cdot A$ and $A \cdot A^T$.

This ensures that singular values are real and non-negative.

=> Definition/Formulas of SVD:

-The Singular Value Decomposition (SVD) of a matrix A is a factorization expressed as:

$$A = U\Sigma V^T,$$

where:

A is an m × n (real or complex).

U is an m × m (or unitary) matrix whose columns are the left singular vectors of A.

Σ is an m × n diagonal matrix with non-negative singular values $\sigma_1, \sigma_2, \ldots, \sigma_r$ on the diagonal [r=min (m, n)].

V^T is the transpose (or conjugate transpose) of an n × n orthogonal (or unitary) matrix V, whose columns are the right singular vectors of A.

-Important Characteristics:

Columns of U: Eigenvectors of $A \cdot A^T$.

Columns of V: Eigenvectors of $A^T \cdot A$.

Singular Values σ_i: Square roots of eigenvalues of $A^T \cdot A$ or $A \cdot A^T$.

Rank of A: Number of non-zero singular values in Σ.

=> At least one theorem related to a SVD finding:

-Theorem (SVD Existence):

For any m × n matrix A, the Singular Value Decomposition (SVD) exists and can be computed as $A = U\Sigma V^T$, where:

-U and V are orthogonal matrices,

-Σ is a diagonal matrix containing the singular values of A, and

-The singular values are the square roots of the eigenvalues of $A^T \cdot A$ or $A^T \cdot A$.

This theorem guarantees that any matrix can be decomposed into orthogonal matrices and a diagonal matrix containing singular values, which are critical in applications like dimensionality reduction and noise filtering.

=> Steps for finding SVD:

1.Compute $A^T \cdot A$ and $A \cdot A^T$:

-Start by calculating the matrices $A^T \cdot A$ (for right singular vectors) and $A \cdot A^T$ (for left singular vectors).

2.Find Eigenvalues and Eigenvectors and U, V:

-Compute the eigenvalues and eigenvectors of $A^T \cdot A$ to determine V.

-Compute the eigenvalues and eigenvectors of $A \cdot A^T$ to determine U.

-Find the matrix U is also an orthogonal matrix that orthogonally diagonalizes $A^T \cdot A$.

-Find the matrix V is also an orthogonal matrix that orthogonally diagonalizes $A \cdot A^T$.

3.Construct Σ:

-The diagonal entries of Σ are the square roots of the eigenvalues of $A^T \cdot A$, arranged in descending order.

4.Form the SVD:

-Combine the matrices U, Σ, and V^T to obtain the SVD of A and the form is: $A=U\Sigma V^T$.

=> Five applications of SVD:

1.Data Compression: SVD helps compress large datasets by retaining only the most significant singular values, reducing the dimensionality while preserving essential features.

2.Image Compression: SVD reduces the amount of data required to store images by approximating them using fewer singular values.

3.Pseudo-Inverse Calculation: SVD allows the calculation of the Moore-Penrose pseudo-inverse for solving overdetermined or underdetermined systems of linear equations.

4.Noise Reduction: By eliminating small singular values, SVD can filter out noise from data.

5.Latent Semantic Analysis (LSA): SVD is used in natural language processing to uncover relationships between terms and documents, enhancing information retrieval and text analysis.

2.2 Question 2

=>Theorem:

Given a square matrix **A**, we wanted to know if it is possible to find an invertible matrix **P** such that:

$$\mathbf{P^{-1}AP = D}$$ (a diagonal matrix)

A square matrix A is called **diagonalizable** if there exists an invertible matrix P such that $\mathbf{P^{-1}AP}$ is a diagonal matrix. In here, matrix P is said to **diagonalize** A.

=>Steps to Diagonalize a Matrix:

1. **Find characteristic polynomial** det $(\lambda I - A)= 0$ and solve for eigenvalues λ.
2. **Find eigenvectors** by solving $(\lambda I - A) \cdot v = 0$ for each λ.
3. **Check for diagonalizability**:

A is diagonalizable if it has n linearly independent eigenvectors.

4. **Construct P and D**:

P: eigenvectors as columns.

D: eigenvalues on the diagonal.

=>Condition:

-A is diagonalizable if the dimension of eigenspaces equals the multiplicity of eigenvalues.

2.3 Question 3

=>Theorem to Construct an Orthonormal Basis (Gram-Schmidt Process):

Let $\{u_1, u_2, \dots, u_k\}$ be a basis for a vector space V.

Let $v_1 = u_1$;

$v_2 = u_2 - \frac{u_2 \cdot v_1}{\|v_1\|^2} v_1$; $\quad v_3 = u_3 - (\frac{u_3 \cdot v_1}{\|v_1\|^2} v_1 + \frac{u_3 \cdot v_2}{\|v_2\|^2} v_2)$;

.

.

.

$v_k = u_k - (\frac{u_k \cdot v_1}{\|v_1\|^2} v_1 + \frac{u_k \cdot v_2}{\|v_2\|^2} v_2 + \cdots + \frac{u_k \cdot v_{k-1}}{\|v_{k-1}\|^2} v_{k-1})$;

Then $\{v_1, v_2, \dots, v_k\}$ is an orthogonal basis for V.

Let $v_1 = u_1$;

$v_2 = u_2 - \frac{u_2 \cdot v_1}{\|v_1\|^2} v_1$; $\quad v_3 = u_3 - (\frac{u_3 \cdot v_1}{\|v_1\|^2} v_1 + \frac{u_3 \cdot v_2}{\|v_2\|^2} v_2)$;

.

.

.

$v_k = u_k - (\frac{u_k \cdot v_1}{\|v_1\|^2} v_1 + \frac{u_k \cdot v_2}{\|v_2\|^2} v_2 + \cdots + \frac{u_k \cdot v_{k-1}}{\|v_{k-1}\|^2} v_{k-1})$;

=>$\{\frac{1}{\|v_1\|} v_1, \frac{1}{\|v_2\|} v_2, \dots, \frac{1}{\|v_k\|} v_k\}$ is an orthonormal basis for V.

2.4 Question 4

=>Theorem:

***)** $S = \{u_1 . u_2, \dots, u_k\}$ and $T = \{v_1, v_2, \dots, v_k\}$ are both bases for $\mathbb{R}^k$.

To find the transition matrix **from S to T**:

Write each vector **from S** in terms of vectors in T.

That is, find $[u_1]_T, [u_2]_T$ and $[u_k]_T$.

So, the transition matrix from S to T is

$$P = ([u_1]_T \quad [u_2]_T \quad [u_k]_T)$$

*) Let S and T be two bases of a vector space and let **P** be the transition matrix from S to T. Then

1) **P** is invertible; and
2) $\mathbf{P}^{-1}$ is the transition matrix from T to S.

Idea:

1) let **Q** be the transition matrix from T to S;
2) Show that **QP** = I (that is, **Q** and **P** are inverses of each other).

=>Conclusion: $Q_{\mathrm{T}\to S} = P^{-1}_{\mathrm{S}\to T}$

=>Step by Step for question 4:

In the problem, we have three vectors $\varepsilon_1, \varepsilon_2, \varepsilon_3$ and each is expressed as a **linear combination** of vectors from two different bases:

$$\varepsilon_1 = \sum_{i=1}^{3} a_i \theta_i, \qquad \varepsilon_2 = \sum_{i=1}^{3} b_i \theta_i, \qquad \varepsilon_3 = \sum_{i=1}^{3} c_i \theta_i$$

-Here:

a_i, b_i, c_i are the coefficients of the linear combinations,

θ, ε represent different bases.

=>Transition Matrix Between Bases

-To transition from one basis to another, we use **transition matrices**:

1. **From Basis ε to Basis θ**:

   -The transition matrix $P_{\varepsilon\to\theta}$ expresses the basis ε vectors as a linear combination of the basis θ vectors.

2. **From Basis θ to Basis ε**:

   -The transition matrix $Q_{\theta\to\varepsilon}$ converts the coordinates in θ back to ε is also $P^{-1}_{\varepsilon\to\theta}$

CHAPTER 3. DESIGNING AND IMPLEMENTING

3.1 Question 1

Ex 1: Let A = $\begin{pmatrix} \mathbf{2} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{3} & \mathbf{0} \end{pmatrix} \Rightarrow$ Rank (A) = 2

∗) We have: $A \cdot A^T = \begin{pmatrix} 4 & 0 \\ 0 & 9 \end{pmatrix}$ is a symmetric matrix

Let $(\lambda I_2 - A \cdot A^T) = \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 4 & 0 \\ 0 & 9 \end{pmatrix}$

$= \begin{pmatrix} \lambda - 4 & 0 \\ 0 & \lambda - 9 \end{pmatrix}$

Characteristic polynomial: $\det(\lambda I_2 - A \cdot A^T) = \begin{vmatrix} \lambda - 4 & 0 \\ 0 & \lambda - 9 \end{vmatrix}$

$= (\lambda - 4)(\lambda - 9)$

Characteristic equation: $\det(\lambda I_2 - A \cdot A^T) = 0 \Rightarrow \begin{bmatrix} \lambda_1 = 4 \\ \lambda_2 = 9 \end{bmatrix}$ are eigenvalues of $A \cdot A^T$

-With $\lambda_1 = 4$. Let $m = \begin{pmatrix} m_1 \\ m_2 \end{pmatrix} \in \mathbb{R}^2 \setminus \{0\}$ is eigenvector of $A \cdot A^T$ that $(4I_2 - A \cdot A^T)m = 0$

Augmented matrix:

$\begin{pmatrix} 0 & 0 \\ 0 & -5 \end{pmatrix} \Rightarrow \begin{cases} m_1 = a \\ m_2 = 0 \end{cases} (a \neq 0) \Rightarrow m = \begin{pmatrix} a \\ 0 \end{pmatrix} = a \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$\Rightarrow E_4 = \left\{ a \begin{pmatrix} 1 \\ 0 \end{pmatrix} \middle| a \neq 0 \right\} = \text{Span} \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$

$\Rightarrow S_4 = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$ is a basis for E_4

-With $\lambda_2 = 9$. Let $n = \begin{pmatrix} n_1 \\ n_2 \end{pmatrix} \in \mathbb{R}^2 \setminus \{0\}$ is eigenvector of $A \cdot A^T$ that $(9I_2 - A \cdot A^T)n = 0$

Augmented matrix:

$\begin{pmatrix} 5 & 0 \\ 0 & 0 \end{pmatrix} \Rightarrow \begin{cases} n_1 = 0 \\ n_2 = b \end{cases} (b \neq 0) \Rightarrow n = \begin{pmatrix} 0 \\ b \end{pmatrix} = b \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$\Rightarrow E_9 = \left\{ b \begin{pmatrix} 0 \\ 1 \end{pmatrix} \middle| b \neq 0 \right\} = \text{Span} \left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$

$\Rightarrow S_9 = \left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$ is a basis for E_9

$S = S_4 \cup S_9 = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$

We already have m·n = 1.0 + 0.1 = 0 $\Rightarrow v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$\|v_1\| = \sqrt{1^2 + 0^2} = 1$

$\|v_2\| = \sqrt{0^2 + 1^2} = 1$

$\Rightarrow w_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, w_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

So that $U = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ is an orthogonal matrix that will orthogonally diagonalize $A \cdot A^T$

$*$) We have: $A^T \cdot A = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ is a synmetic matrix

Characteristics polynomial: $\det(\lambda I_3 - A^T \cdot A) = \begin{vmatrix} \lambda - 4 & 0 & 0 \\ 0 & \lambda - 9 & 0 \\ 0 & 0 & \lambda \end{vmatrix}$

$= (\lambda - 4)(\lambda - 9)\lambda$

Characteristic equation: $\det(\lambda I_3 - A^T \cdot A) = 0 \Rightarrow \begin{bmatrix} \lambda_1 = 4 \\ \lambda_2 = 9 \\ \lambda_3 = 0 \end{bmatrix}$ are eigenvalues of $A^T \cdot A$

-With $\lambda_1 = 4$. Let $m = \begin{pmatrix} m_1 \\ m_2 \\ m_3 \end{pmatrix} \in \mathbb{R}^3 \setminus \{0\}$ is eigenvector of $A^T \cdot A$ that $(4I_3 - A^T \cdot A)m = 0$

Augmented matrix:

$\begin{pmatrix} 0 & 0 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & 4 \end{pmatrix} \Rightarrow \begin{cases} m_1 = a \\ m_2 = 0 \\ m_3 = 0 \end{cases} (a \neq 0) \Rightarrow m = \begin{pmatrix} a \\ 0 \\ 0 \end{pmatrix} = a \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

$\Rightarrow E_4 = \left\{ a \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \middle| a \neq 0 \right\} = \text{Span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\}$

$\Rightarrow S_4 = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\}$ is a basis for E_4

-With $\lambda_2 = 9$. Let $n = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} \in \mathbb{R}^3 \setminus \{0\}$ is eigenvector of $A^T \cdot A$ that $(9I_3 - A^T \cdot A)n = 0$

Augmented matrix:

$\begin{pmatrix} 4 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 9 \end{pmatrix} \Rightarrow \begin{cases} n_1 = 0 \\ n_2 = b \ (a \neq 0) \\ n_3 = 0 \end{cases} \Rightarrow n = \begin{pmatrix} 0 \\ b \\ 0 \end{pmatrix} = b \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$

$\Rightarrow E_9 = \left\{ b \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \middle| b \neq 0 \right\} = \text{Span} \left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$

$\Rightarrow S_9 = \left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$ is a basis for E_9

-With $\lambda_3 = 0$. Let $p = \begin{pmatrix} p_1 \\ p_2 \\ p_3 \end{pmatrix} \in \mathbb{R}^3 \setminus \{0\}$ is eigenvector of $A^T \cdot A$ that $(0I_3 - A^T \cdot A)p = 0$

Augmented matrix:

$$\begin{pmatrix} -4 & 0 & 0 \\ 0 & -9 & 0 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \begin{cases} p_1 = 0 \\ p_2 = 0 \ (c \neq 0) \\ p_3 = c \end{cases} \Rightarrow p = \begin{pmatrix} 0 \\ 0 \\ c \end{pmatrix} = c \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$\Rightarrow E_0 = \left\{ c \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \middle| c \neq 0 \right\} = \text{Span} \left\{ \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$

$\Rightarrow S_0 = \left\{ \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$ is a basis for E_0

We have $S = S_4 \cup S_9 \cup S_0 = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$

We already have:

$m \cdot n = 1 \cdot 0 + 0 \cdot 1 + 0 \cdot 0 = 0$

$n \cdot p = 0 \cdot 0 + 1 \cdot 0 + 0 \cdot 1 = 0$

$m \cdot p = 1 \cdot 0 + 0 \cdot 0 + 0 \cdot 1 = 0$

$\Rightarrow v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, v_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, v_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

$\Rightarrow \|v_1\| = \sqrt{1^2 + 0^2 + 0^2} = 1$

$\|v_2\| = \sqrt{0^2 + 1^2 + 0^2} = 1$

$\|v_3\| = \sqrt{0^2 + 0^2 + 1^2} = 1$

$\Rightarrow w_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, w_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, w_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

So that V= $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ is an orthogonal matrix that will orthogonally diagonalize

$A^T \cdot A$ and $\Sigma = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \end{pmatrix}$

Thus, we have SVD: $A = U \cdot \Sigma \cdot V^T = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}^T$

Ex 2: Let A = $\begin{pmatrix} \mathbf{2} & \mathbf{0} & \mathbf{0} \\ \mathbf{2} & \mathbf{0} & \mathbf{0} \end{pmatrix} \xrightarrow{R_2 - R_1 \to R_2} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow$ Rank (A) = 1

∗) We have: $A \cdot A^T = \begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}$ is a symmetric matrix

Let $(\lambda I_2 - A \cdot A^T) = \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}$

$= \begin{pmatrix} \lambda - 4 & -4 \\ -4 & \lambda - 4 \end{pmatrix}$

Characteristic polynomial: $\det(\lambda I_2 - A \cdot A^T) = \begin{vmatrix} \lambda - 4 & -4 \\ -4 & \lambda - 4 \end{vmatrix}$

$= (\lambda - 4)^2 - 16$

$= (\lambda - 8)\lambda$

Characteristic equation: $\det(\lambda I_2 - A \cdot A^T) = 0 \Rightarrow \begin{bmatrix} \lambda_1 = 8 \\ \lambda_2 = 0 \end{bmatrix}$ are eigenvalues of $A \cdot A^T$

-With $\lambda_1 = 8$. Let $m = \begin{pmatrix} m_1 \\ m_2 \end{pmatrix} \in \mathbb{R}^2 \setminus \{0\}$ is eigenvector of $A \cdot A^T$ that $(8I_2 - A \cdot A^T)m = 0$

Augmented matrix:

$\begin{pmatrix} 4 & -4 \\ -4 & 4 \end{pmatrix} \xrightarrow{R_2 + R_1 \to R_2} \begin{pmatrix} 4 & -4 \\ 0 & 0 \end{pmatrix} \Rightarrow \begin{cases} m_1 = a \\ m_2 = a \end{cases} (a \neq 0) \Rightarrow m = \begin{pmatrix} a \\ a \end{pmatrix} = a \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$\Rightarrow E_8 = \left\{ a \begin{pmatrix} 1 \\ 1 \end{pmatrix} \middle| a \neq 0 \right\} = \text{Span} \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$

$\Rightarrow S_8 = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$ is a basis for E_8

-With $\lambda_2 = 0$. Let $n = \begin{pmatrix} n_1 \\ n_2 \end{pmatrix} \in \mathbb{R}^2 \setminus \{0\}$ is eigenvector of $A \cdot A^T$ that $(0I_2 - A \cdot A^T)n = 0$

Augmented matrix:

$\begin{pmatrix} -4 & -4 \\ -4 & -4 \end{pmatrix} \xrightarrow{R_2 - R_1 \to R_2} \begin{pmatrix} -4 & -4 \\ 0 & 0 \end{pmatrix} \Rightarrow \begin{cases} n_1 = -b \\ n_2 = b \end{cases} (b \neq 0) \Rightarrow n = \begin{pmatrix} -b \\ b \end{pmatrix} = b \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

$\Rightarrow E_0 = \left\{ b \begin{pmatrix} -1 \\ 1 \end{pmatrix} \middle| b \neq 0 \right\} = \text{Span} \left\{ \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\}$

$\Rightarrow S_0 = \left\{ \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\}$ is a basis for E_0

Let $S = S_8 \cup S_0 = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$

We already have $m \cdot n = 1.1 + (-1).1 = 0 \Rightarrow v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $v_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

$\Rightarrow \|v_1\| = \sqrt{1^2 + 1^2} = \sqrt{2}$

$\|v_2\| = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$

$\Rightarrow w_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$w_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

So that $U = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$ is an orthogonal matrix that will orthogonally diagonalize $A \cdot A^T$

$*$) We have: $A^T \cdot A = \begin{pmatrix} 8 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ is a synmetic matrix

Let $(\lambda I_3 - A^T \cdot A) = \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} - \begin{pmatrix} 8 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} \lambda - 8 & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix}$

Characteristics polynomial: $\det(\lambda I_3 - A^T \cdot A) = \begin{vmatrix} \lambda - 8 & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{vmatrix}$

$= (\lambda - 8)\lambda^2$

Characteristic equation: $\det(\lambda I_3 - A^T \cdot A) = 0 \Rightarrow \begin{bmatrix} \lambda_1 = 8 \\ \lambda_2 = 0 \end{bmatrix}$ are eigenvalues of $A^T \cdot A$

-With $\lambda_1 = 8$. Let $m = \begin{pmatrix} m_1 \\ m_2 \\ m_3 \end{pmatrix} \in \mathbb{R}^3 \setminus \{0\}$ is eigenvector of $A^T \cdot A$ that $(8I_3 - A^T \cdot A)m = 0$

Augmented matrix:

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 8 \end{pmatrix} \Rightarrow \begin{cases} m_1 = a \\ m_2 = 0 \\ m_3 = 0 \end{cases} (a \neq 0) \Rightarrow m = \begin{pmatrix} a \\ 0 \\ 0 \end{pmatrix} = a \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \mathrm{E}_8 = \left\{ a \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \middle| a \neq 0 \right\} = \mathrm{Span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\}$$

$$\Rightarrow \mathrm{S}_8 = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\} \text{ is a basis for } \mathrm{E}_8$$

-With $\lambda_2 = 0$. Let $n = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} \in \mathbb{R}^3 \setminus \{0\}$ is eigenvector of $\mathrm{A}^T \cdot \mathrm{A}$ that $(0\mathrm{I}_3 - \mathrm{A}^T \cdot \mathrm{A})n = 0$

Augmented matrix:

$$\begin{pmatrix} -8 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \begin{cases} n_1 = 0 \\ n_2 = b \\ n_3 = \mathrm{c} \end{cases} (b, c \neq 0) \Rightarrow n = \begin{pmatrix} 0 \\ b \\ c \end{pmatrix} = b \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \mathrm{c} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\Rightarrow \mathrm{E}_0 = \left\{ b \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \mathrm{c} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \middle| b, c \neq 0 \right\} = \mathrm{Span} \left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

$$\Rightarrow \mathrm{S}_0 = \left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\} \text{ is a basis for } \mathrm{E}_0$$

We have $\mathrm{S} = \mathrm{S}_8 \cup \mathrm{S}_0 = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$

We already have:

$\mathrm{m} \cdot \mathrm{n}_1 = 1 \cdot 0 + 0 \cdot 1 + 0 \cdot 0 = 0$

$\mathrm{n}_1 \cdot \mathrm{n}_2 = 0 \cdot 0 + 1 \cdot 0 + 0 \cdot 1 = 0$

$\mathrm{n}_2 \cdot \mathrm{m} = 0 \cdot 1 + 0 \cdot 0 + 0 \cdot 1 = 0$

$$\Rightarrow v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, v_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, v_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\Rightarrow \|v_1\| = \sqrt{1^2 + 0^2 + 0^2} = 1$$

$$\|v_2\| = \sqrt{0^2 + 1^2 + 0^2} = 1$$

$$\|v_3\| = \sqrt{0^2 + 0^2 + 1^2} = 1$$

$\Rightarrow w_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, w_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, w_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

So that $V = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ is an orthogonal matrix that will orthogonally diagonalize $A^T \cdot A$ and $\Sigma = \begin{pmatrix} 2\sqrt{2} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

Thus, we have SVD: $A = U \cdot \Sigma \cdot V^T = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \cdot \begin{pmatrix} 2\sqrt{2} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}^T$

3.2 Question 2

We have:

$$(\lambda I_3 - A) = \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \lambda - 1 & -2 & -2 \\ -2 & \lambda - 1 & -1 \\ 0 & 0 & \lambda - 1 \end{pmatrix}$$

Characteristic polynomial: $\det(\lambda I_3 - A) = \begin{vmatrix} \lambda - 1 & -2 & -2 \\ -2 & \lambda - 1 & -1 \\ 0 & 0 & \lambda - 1 \end{vmatrix}$

$$= (\lambda - 1)^3 - 4(\lambda - 1)$$

$$= (\lambda^2 - 2\lambda - 3)(\lambda - 1)$$

Characteristic equation: $\det(\lambda I_3 - A) = 0$

$$\Rightarrow (\lambda^2 - \lambda - 3)(\lambda - 1) = 0$$

$$\Rightarrow \begin{cases} \lambda_1 = 1 \\ \lambda_2 = 3 \\ \lambda_3 = -1 \end{cases} \text{ are eigenvalues of A}$$

–With $\lambda_1 = 1$. Let $m = \begin{pmatrix} m_1 \\ m_2 \\ m_3 \end{pmatrix} \in \mathbb{R}^3 \setminus \{0\}$ is eigenvector of A that $(1I_3 - A)m = 0$

Augmented matrix:

$$\begin{pmatrix} 0 & -2 & -2 \\ -2 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \begin{cases} -2m_2 - 2m_3 = 0 \\ -2m_1 - m_3 = 0 \end{cases} \Rightarrow \begin{cases} m_1 = a \\ m_2 = 2a \\ m_3 = -2a \end{cases} \quad (a \neq 0)$$

$\Rightarrow m = \begin{pmatrix} a \\ 2a \\ -2a \end{pmatrix} = a\begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$

$\Rightarrow \mathrm{E}_1 = \left\{ a\begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} \middle| a \neq 0 \right\} = \mathrm{Span}\left\{ \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} \right\}$

$\Rightarrow \mathrm{S}_1 = \left\{ \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} \right\}$ is a basis for E_1

−With $\lambda_2 = 3$. Let $n = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} \in \mathbb{R}^3 \setminus \{0\}$ is eigenvector of A that $(3\mathrm{I}_3 - A)n = 0$

Augmented matrix:

$$\begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -1 \\ 0 & 0 & 2 \end{pmatrix} \xrightarrow{C_3 \leftrightarrow C_1} \begin{pmatrix} -2 & -2 & 2 \\ -1 & 2 & -2 \\ 2 & 0 & 0 \end{pmatrix} \xrightarrow[R_3 + R_1 \to R_3]{2R_2 - R_1 \to R_2} \begin{pmatrix} -2 & -2 & 2 \\ 0 & 6 & -6 \\ 0 & -2 & 2 \end{pmatrix}$$

$$\xrightarrow{6R_3 + 2R_2 \to R_3} \begin{pmatrix} -2 & -2 & -2 \\ 0 & 6 & -6 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\Rightarrow \begin{cases} -2n_1 - 2n_2 + 2n_3 = 0 \\ 6n_2 - 6n_3 = 0 \end{cases}$$

$$\Rightarrow \begin{cases} n_1 = 0 \\ n_2 = b \ (b \neq 0) \\ n_3 = b \end{cases} \Rightarrow n = \begin{pmatrix} 0 \\ b \\ b \end{pmatrix} = b\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

$\Rightarrow \mathrm{E}_3 = \left\{ b\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \middle| b \neq 0 \right\} = \mathrm{Span}\left\{ \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}$

$\Rightarrow \mathrm{S}_3 = \left\{ \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}$ is a basis for E_3

−With $\lambda_3 = -1$. Let $p = \begin{pmatrix} p_1 \\ p_2 \\ p_3 \end{pmatrix} \in \mathbb{R}^3 \setminus \{0\}$ is eigenvector of A that $(-1\mathrm{I}_3 - \mathrm{A})p = 0$

Augmented matrix:

$$\begin{pmatrix} -2 & -2 & -2 \\ -2 & -2 & -1 \\ 0 & 0 & -2 \end{pmatrix} \xrightarrow{C_3 \leftrightarrow C_1} \begin{pmatrix} -2 & -2 & -2 \\ -1 & -2 & -2 \\ -2 & 0 & 0 \end{pmatrix} \xrightarrow[R_3 - R_1 \to R_3]{2R_2 - R_1 \to R_2} \begin{pmatrix} -2 & -2 & 2 \\ 0 & -2 & -2 \\ 0 & 2 & 2 \end{pmatrix}$$

$$\xrightarrow{R_3 + R_2 \to R_3} \begin{pmatrix} -2 & -2 & -2 \\ 0 & -2 & -2 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\Rightarrow \begin{cases} -2p_1 - 2p_2 - 2p_3 = 0 \\ -2p_2 - 2p_3 = 0 \end{cases} \Rightarrow \begin{cases} p_1 = 0 \\ p_2 = -c \ (c \neq 0) \\ p_3 = c \end{cases} \Rightarrow p = \begin{pmatrix} 0 \\ -c \\ c \end{pmatrix} = c \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$$

$$\Rightarrow \mathrm{E}_{-1} = \left\{ c \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \middle| c \neq 0 \right\} = \mathrm{Span} \left\{ \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \right\}$$

$$\Rightarrow \mathrm{S}_{-1} = \left\{ \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \right\} \text{ is a basis for } \mathrm{E}_{-1}$$

$$\text{Let } \mathrm{S} = \mathrm{S}_1 \cup \mathrm{S}_3 \cup \mathrm{S}_{-1} = \left\{ \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \right\}$$

$|S| = 3$, $\mathrm{A} \in \mathrm{m}_3 \Rightarrow$ A is diagonalizable

With $\mathrm{P} = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & -1 \\ 2 & 1 & 1 \end{pmatrix}$ is a matrix that diagonalizes A and the diagonalization is:

$$\mathrm{P}^{-1}\mathrm{AP} = \mathrm{D} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

3.3 Question 3

We already have: $\mathrm{U}_1 \cdot \mathrm{U}_2 = 4 \cdot 8 + 8 \cdot (-8) + 8 \cdot 4 = 0$

$$\mathrm{U}_2 \cdot \mathrm{U}_3 = 8 \cdot 8 + 8 \cdot (-4) + 4 \cdot (-8) = 0$$

$$\mathrm{U}_1 \cdot \mathrm{U}_3 = 8 \cdot 4 + 4 \cdot 8 + 8 \cdot (-8) = 0$$

$\Rightarrow u_1 = (4, -8, 8)$

$u_2 = (8, 8, 4)$

$u_3 = (8, -4, -8)$

$\Rightarrow \|v_1\| = \sqrt{4^2 + (-8)^2 + 8^2} = 12$

$\|v_2\| = \sqrt{8^2 + 8^2 + 4^2} = 12$

$$\|v_3\| = \sqrt{8^2 + (-4)^2 + (-8)^2} = 12$$

$$\Rightarrow w_1 = \frac{1}{12}(4, -8, 8)$$

$$w_2 = \frac{1}{12}(8, 8, 4)$$

$$w_3 = \frac{1}{12}(8, -4, -8)$$

$$\Rightarrow \text{An orthonormal basis form to } \mathbb{R}^3 \text{ is } W = \{w_1, w_2, w_3\} = \begin{pmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{-2}{3} & \frac{2}{3} & \frac{-1}{3} \\ \frac{2}{3} & \frac{1}{3} & \frac{-2}{3} \end{pmatrix}$$

3.4 Question 4

a) We have linear combination

$$\varepsilon_1 = \sum_{i=1}^{3} a_i \theta_i, \qquad \varepsilon_2 = \sum_{i=1}^{3} b_i \theta_i, \qquad \varepsilon_3 = \sum_{i=1}^{3} c_i \theta_i$$

Augmented matrix:

$$\left(\begin{array}{ccc|c|c|c} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \Rightarrow \begin{cases} a_1 = \frac{1}{2} \\ a_2 = \frac{-1}{2} \\ a_3 = \frac{-1}{2} \end{cases}; \begin{cases} b_1 = \frac{1}{2} \\ b_2 = \frac{1}{2} \\ b_3 = \frac{-1}{2} \end{cases}; \begin{cases} c_1 = \frac{-1}{2} \\ c_2 = \frac{1}{2} \\ c_3 = \frac{1}{2} \end{cases}$$

$$\Rightarrow [\varepsilon_1]_\theta = \begin{pmatrix} \frac{1}{2} \\ \frac{-1}{2} \\ \frac{1}{2} \end{pmatrix}; \ [\varepsilon_2]_\theta = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{-1}{2} \end{pmatrix}; \ [\varepsilon_3]_\theta = \begin{pmatrix} \frac{-1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$$

The transition matrix from ε to θ is:

$$P_{\varepsilon \to \theta} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & \frac{-1}{2} \\ \frac{-1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{-1}{2} & \frac{1}{2} \end{pmatrix}$$

b) The transition matrix from θ to ε is:

$$Q_{\theta \to \varepsilon} = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

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